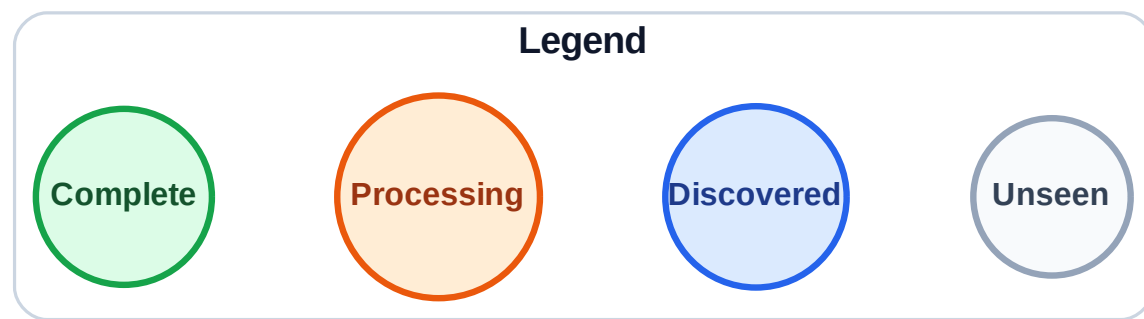
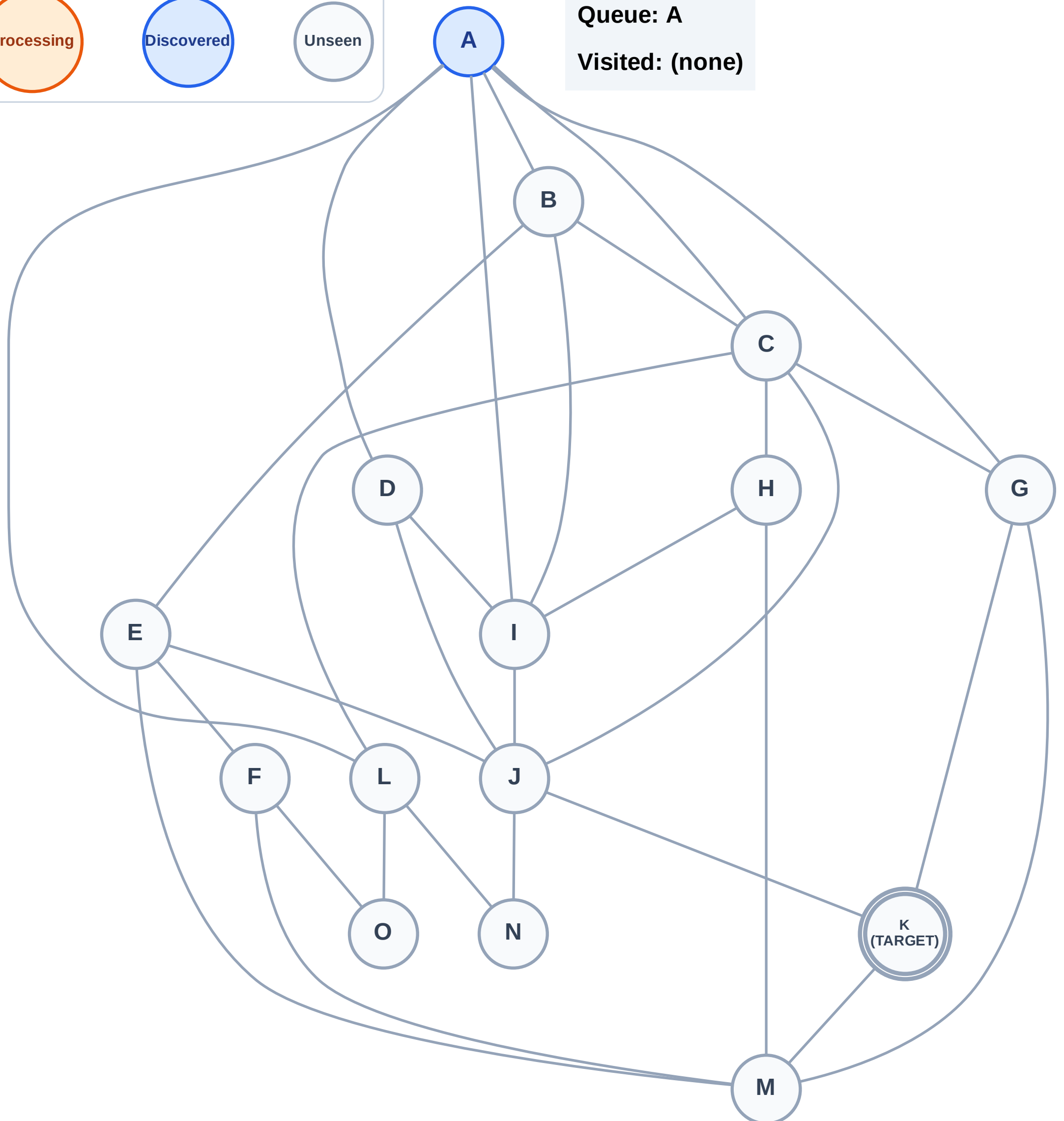


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Queue: A
Visited: (none)



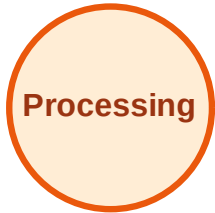
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



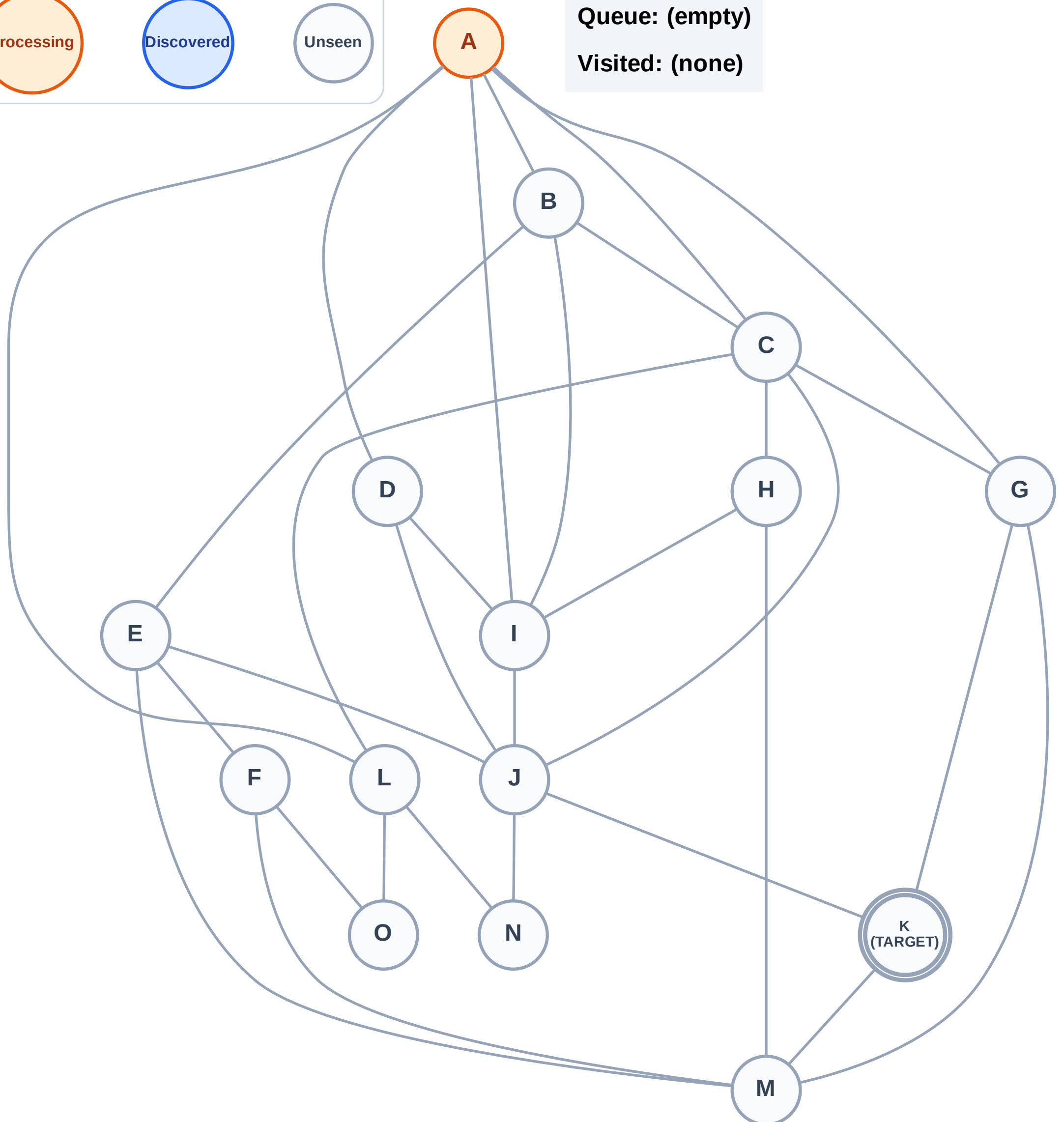
Discovered



Unseen

Queue: (empty)

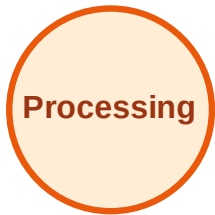
Visited: (none)



BFS Traversal

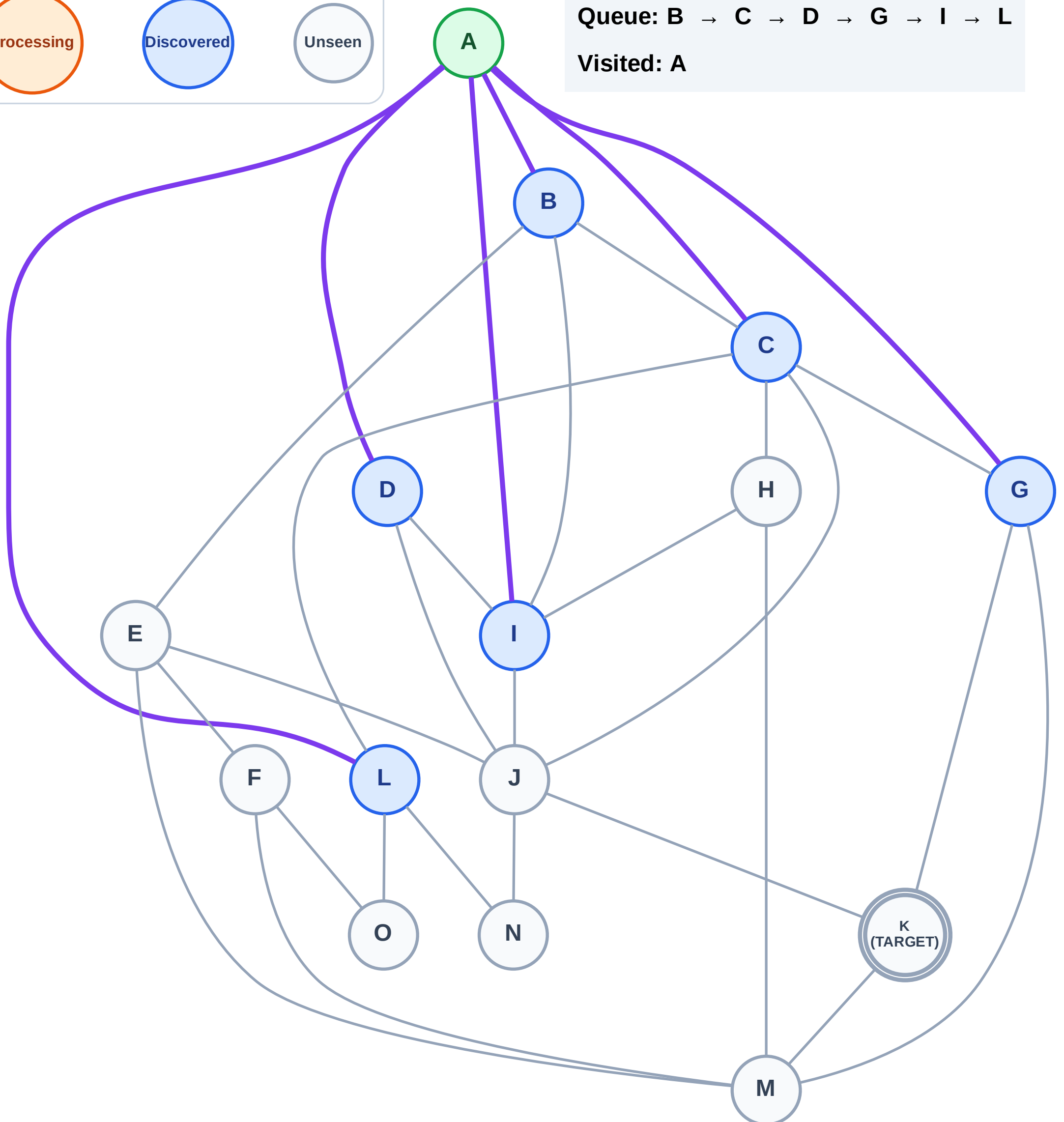
Step 3: Discover B, C, D, G, I, L from A

Legend



Queue: B → C → D → G → I → L

Visited: A



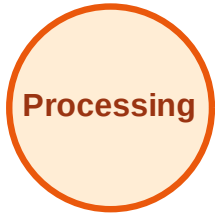
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



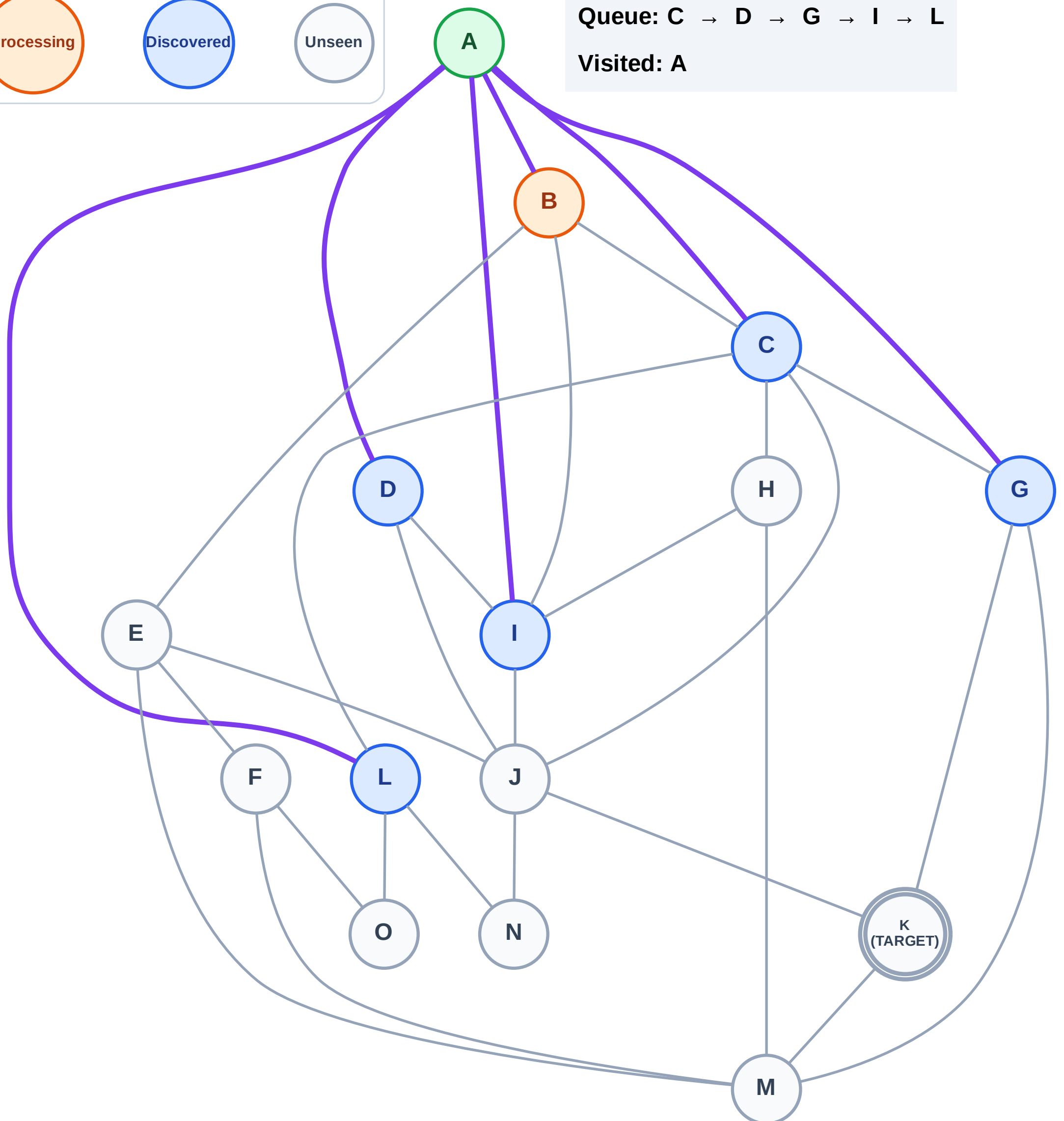
Discovered



Unseen

Queue: C → D → G → I → L

Visited: A



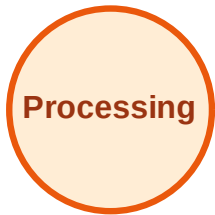
BFS Traversal

Step 5: Discover E from B

Legend



Complete



Processing



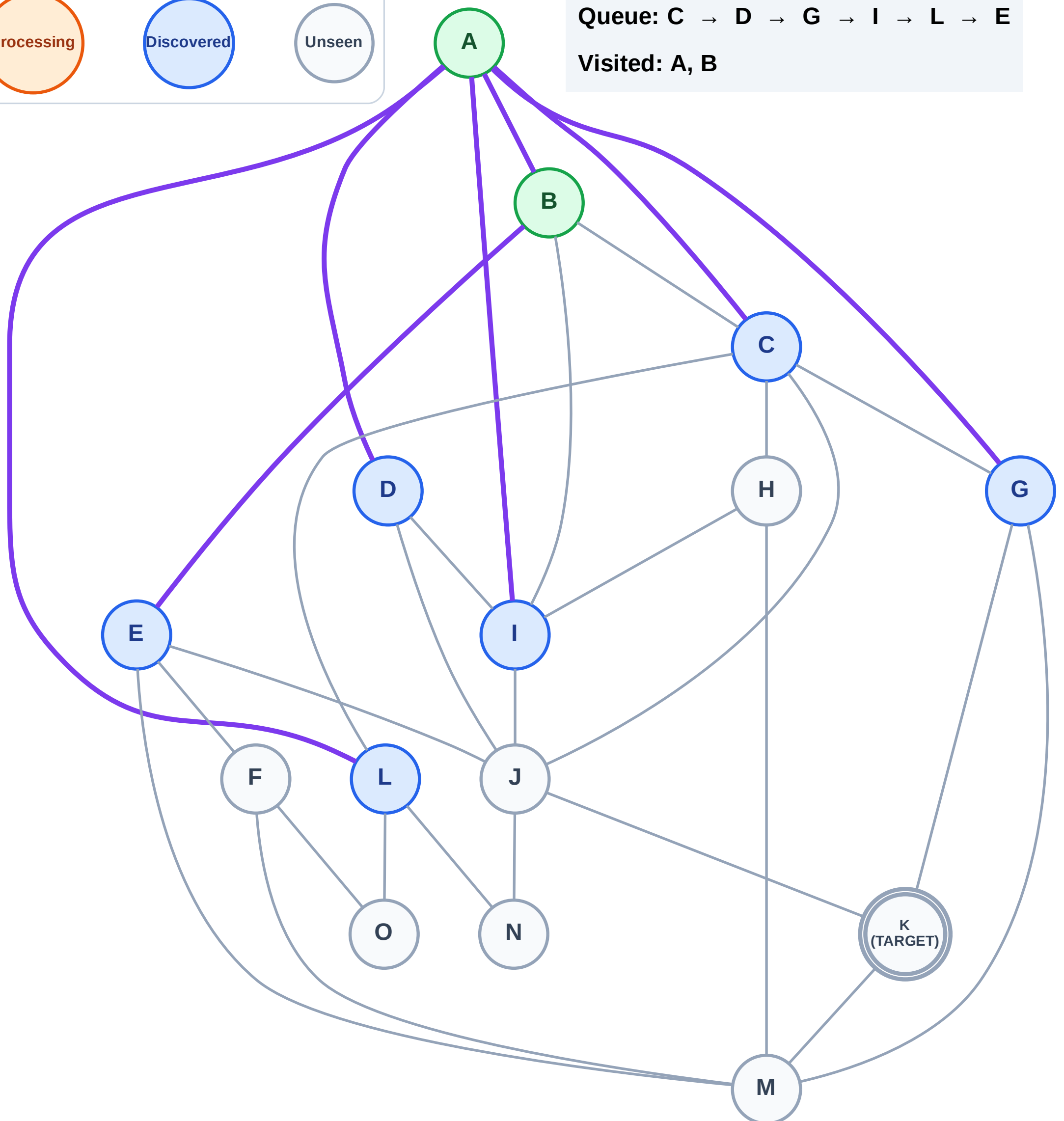
Discovered



Unseen

Queue: C → D → G → I → L → E

Visited: A, B



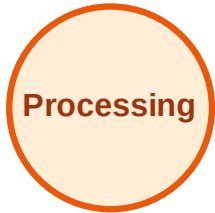
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



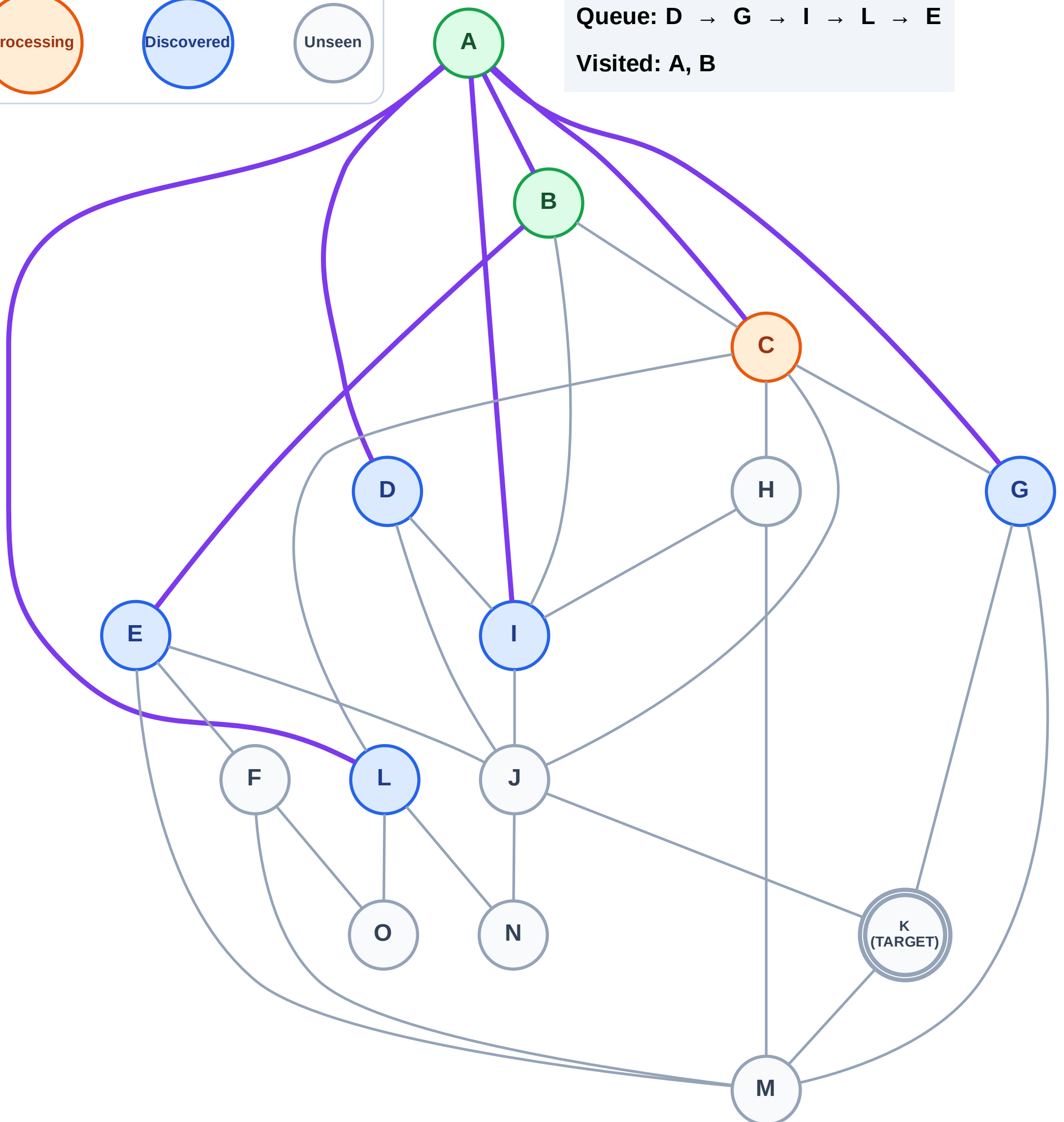
Discovered



Unseen

Queue: D → G → I → L → E

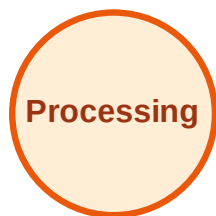
Visited: A, B



BFS Traversal

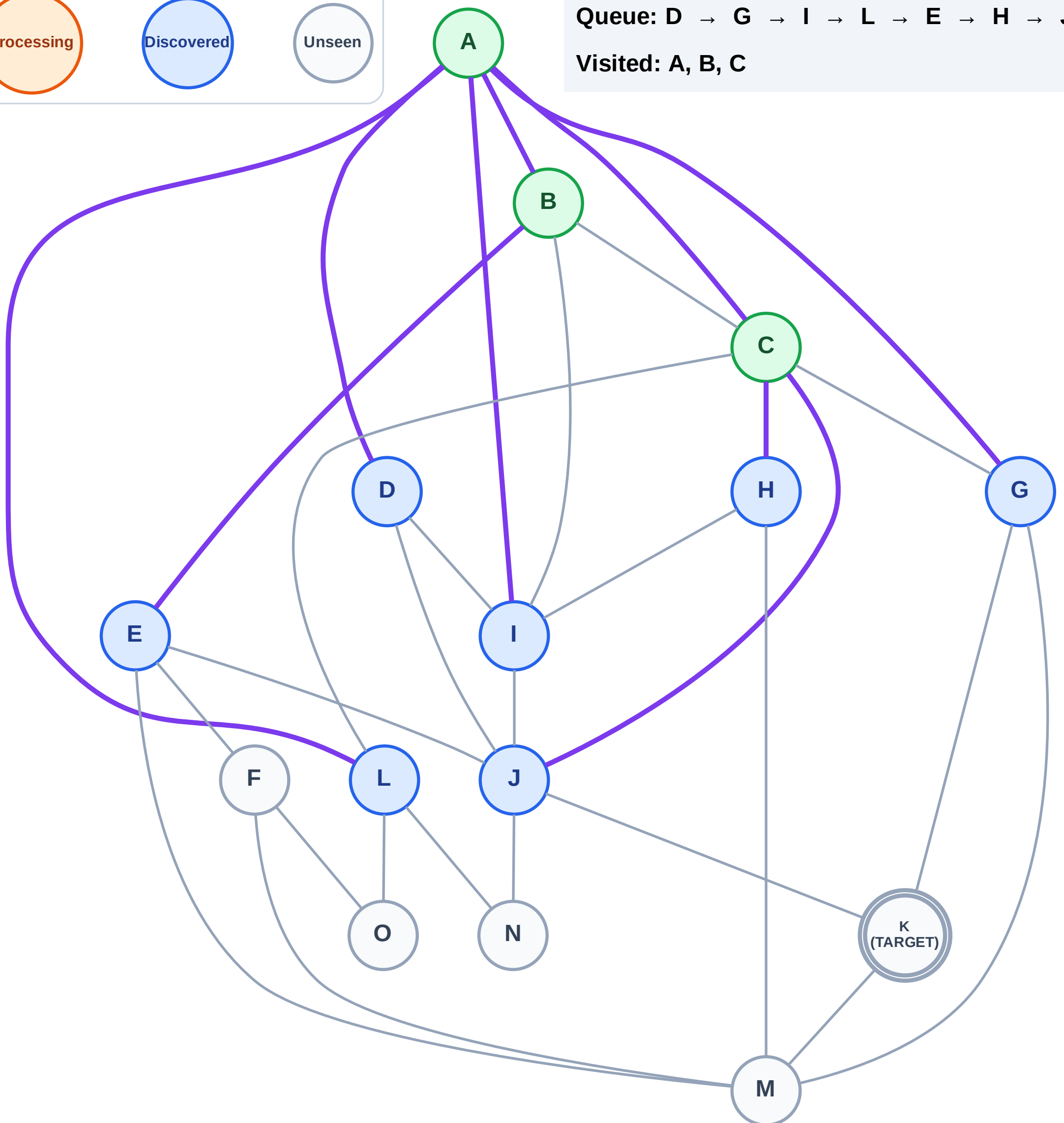
Step 7: Discover H, J from C

Legend



Queue: D → G → I → L → E → H → J

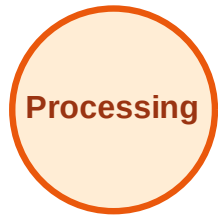
Visited: A, B, C



BFS Traversal

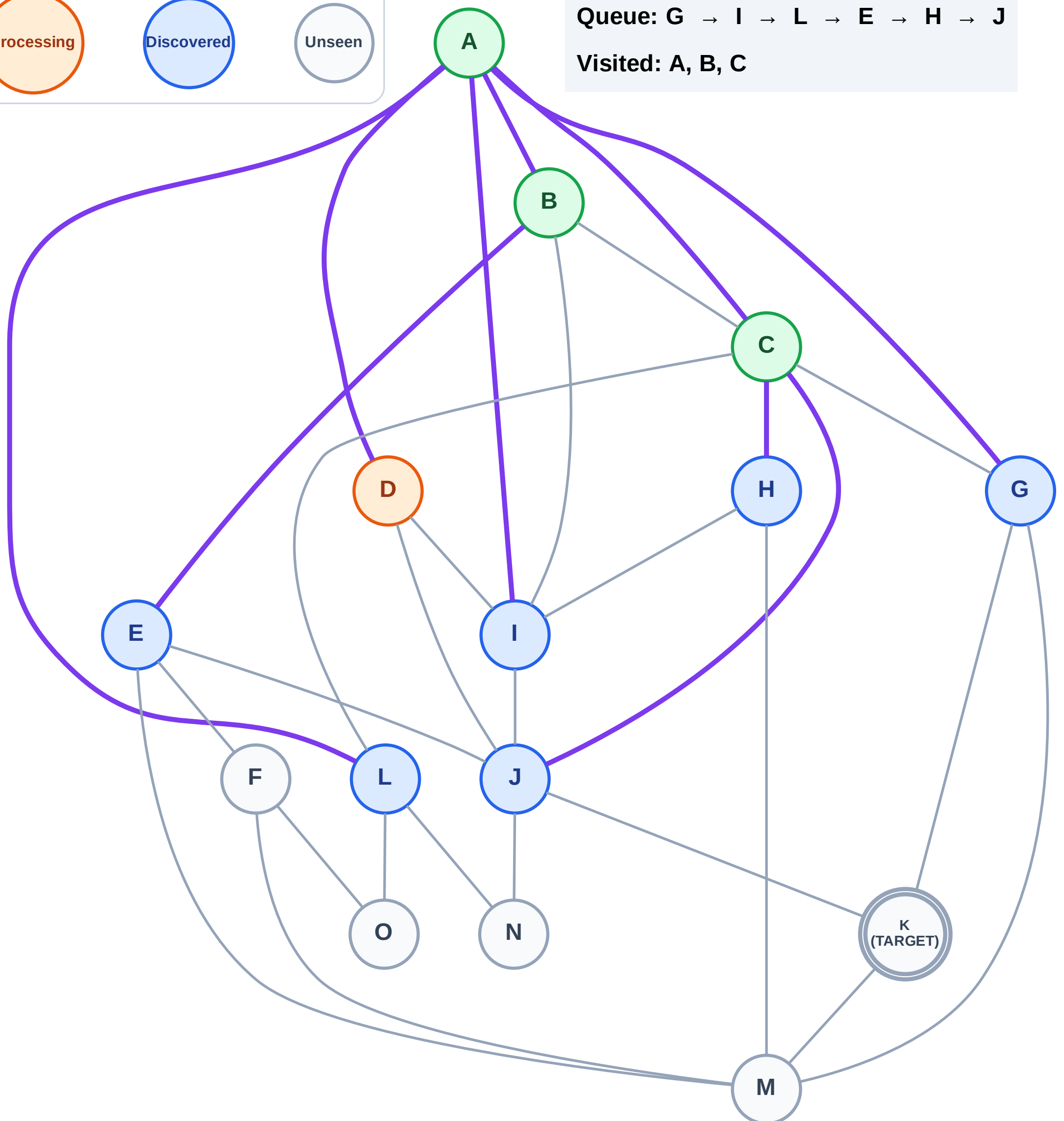
Step 8: Process node D

Legend



Queue: G → I → L → E → H → J

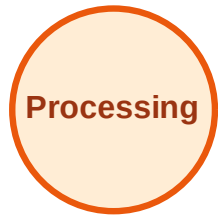
Visited: A, B, C



BFS Traversal

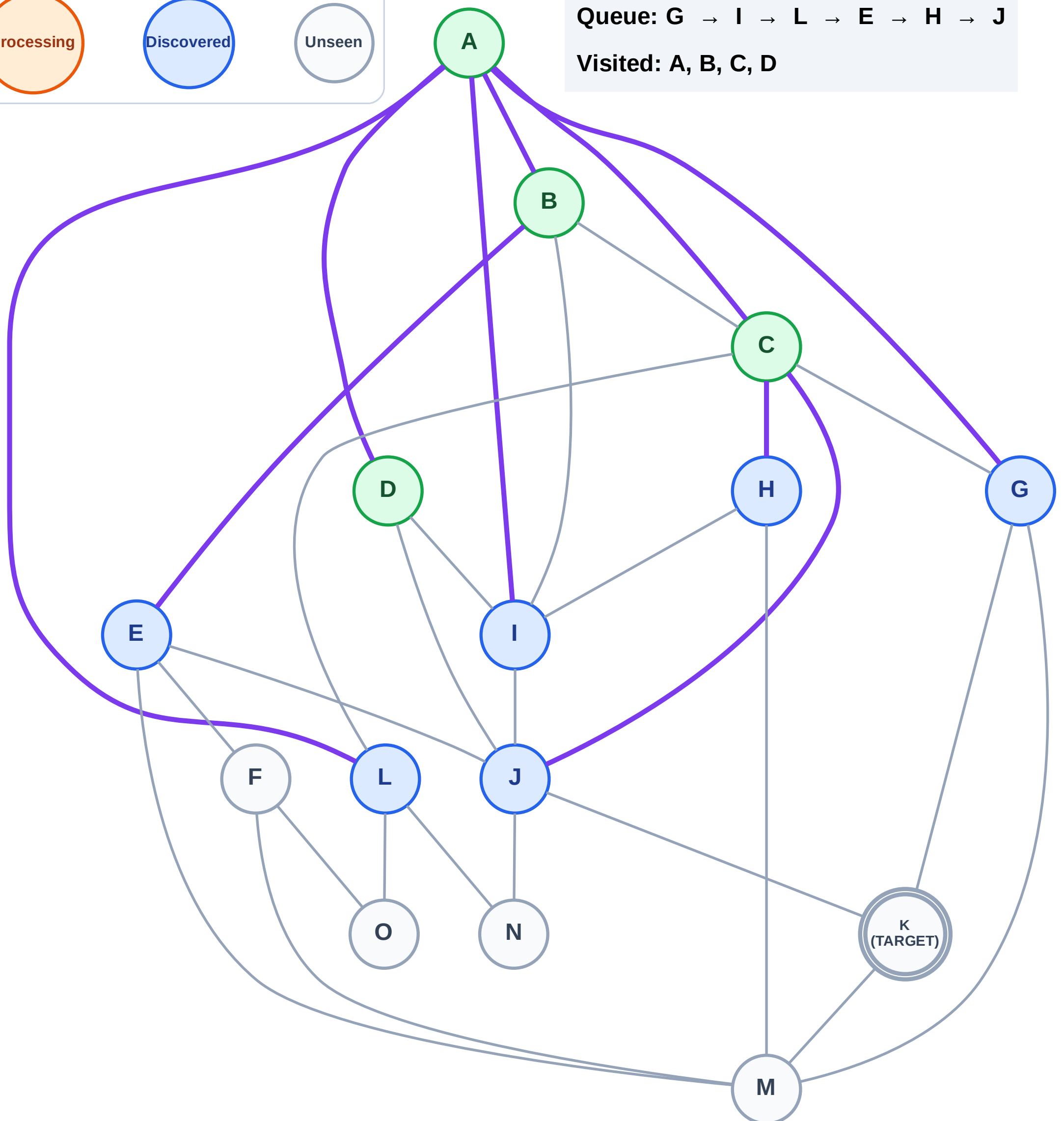
Step 9: No new neighbors from D

Legend



Queue: G → I → L → E → H → J

Visited: A, B, C, D



BFS Traversal

Step 10: Process node G

Legend

Complete

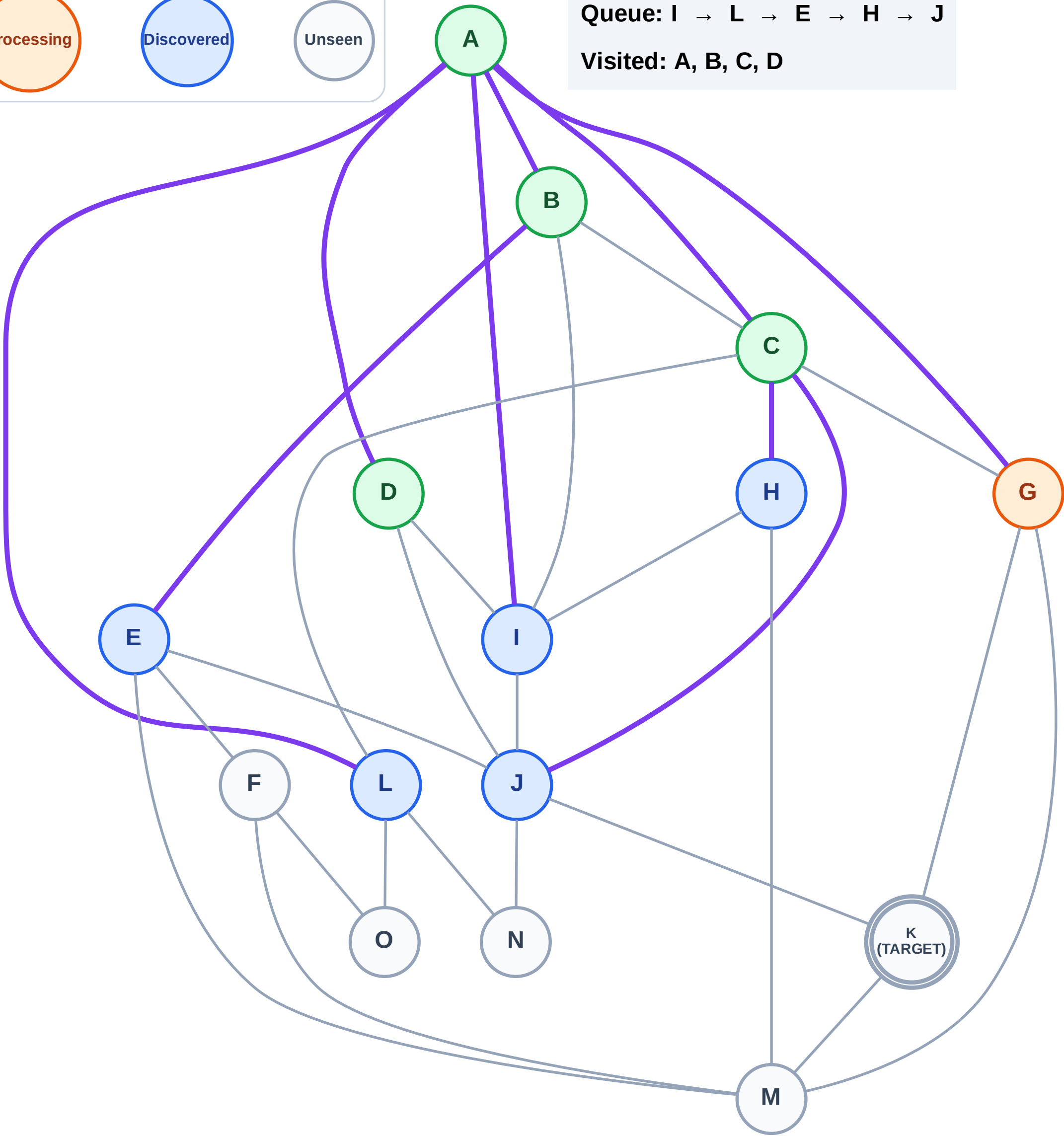
Processing

Discovered

Unseen

Queue: I → L → E → H → J

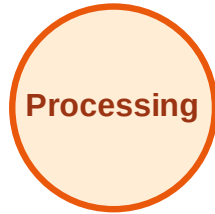
Visited: A, B, C, D



BFS Traversal

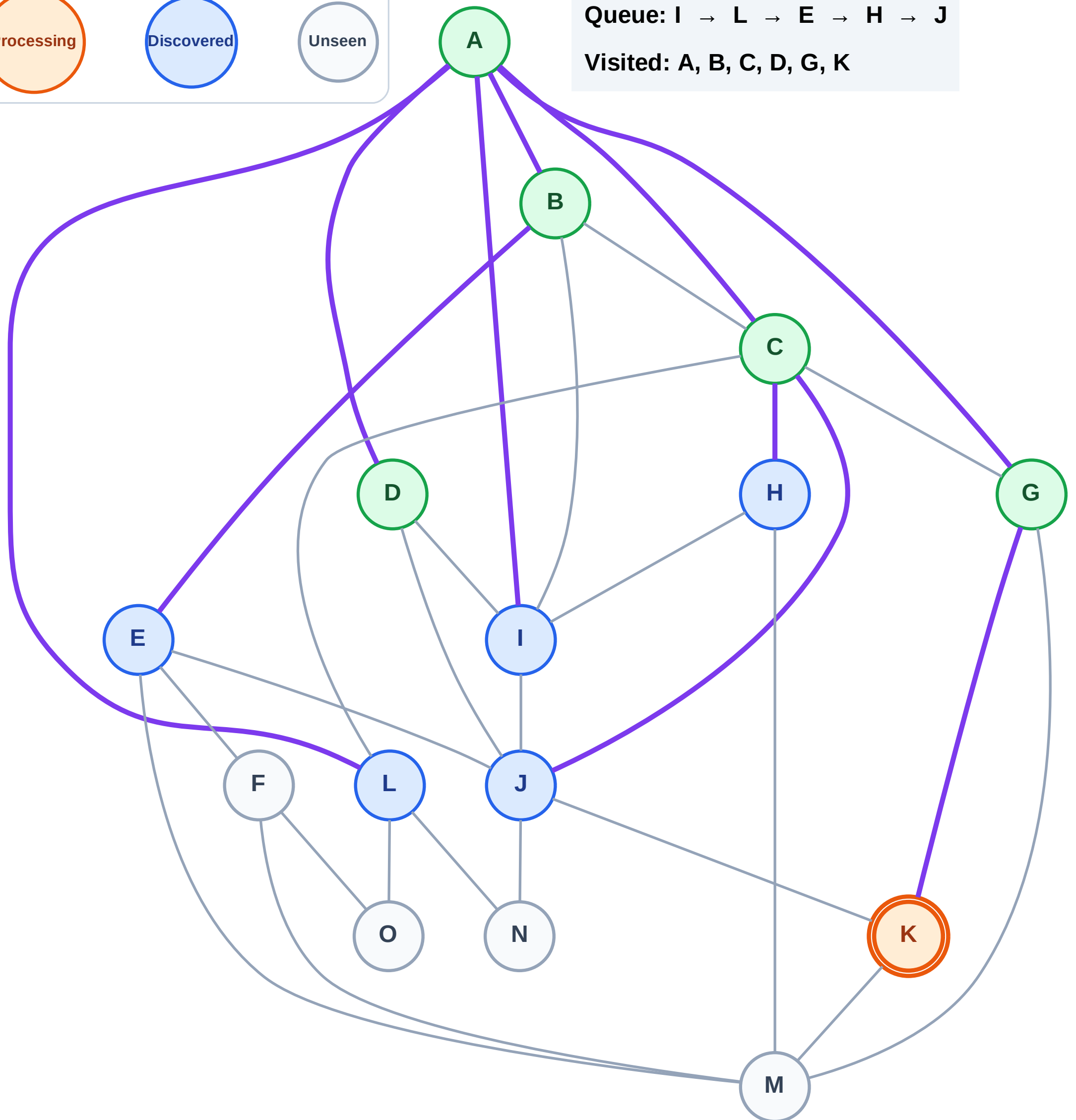
Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → G → K

Legend



Queue: I → L → E → H → J

Visited: A, B, C, D, G, K



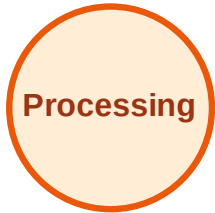
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



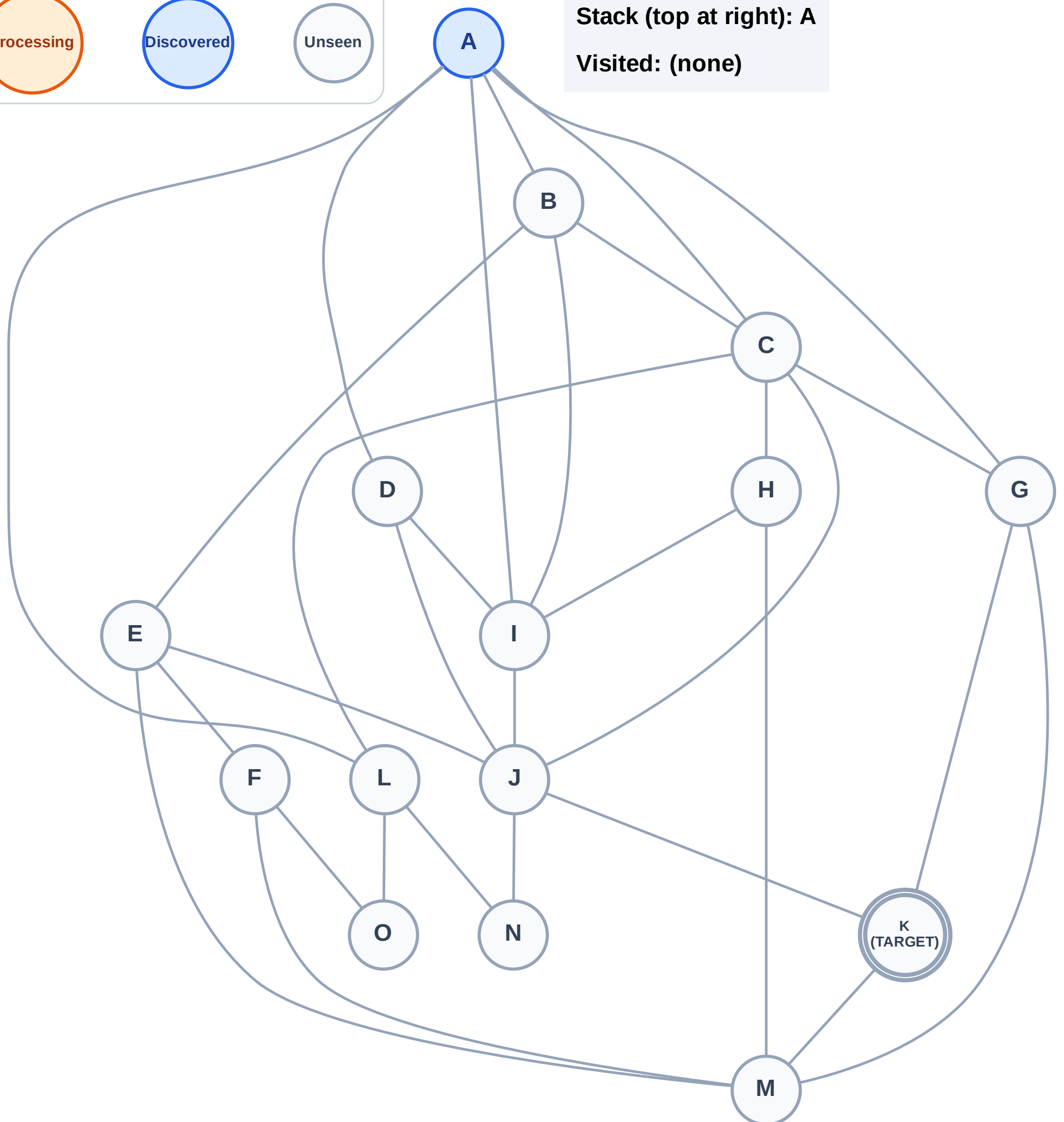
Discovered



Unseen

Stack (top at right): A

Visited: (none)



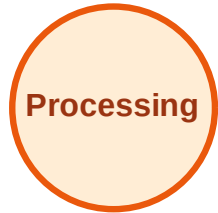
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



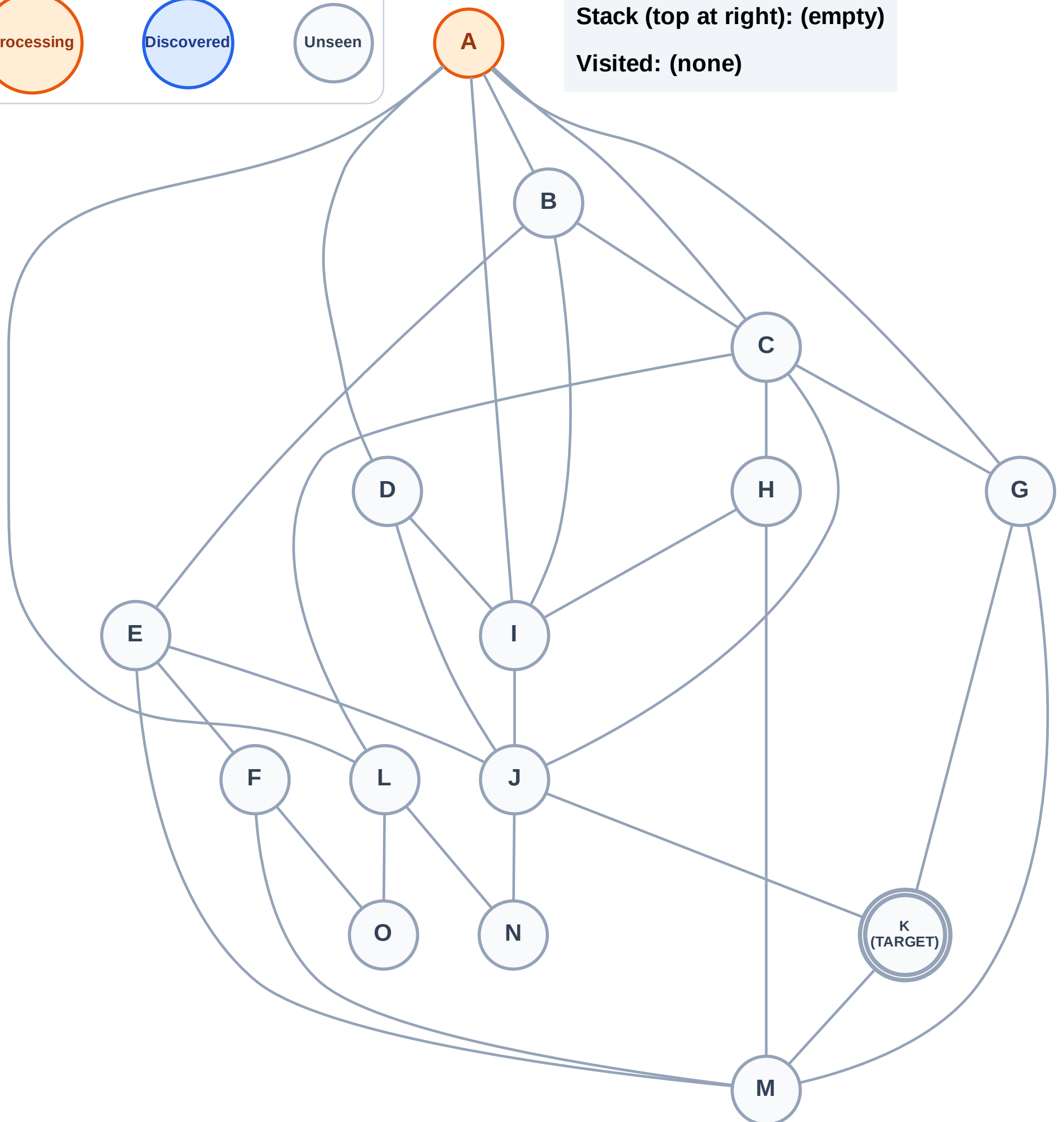
Discovered



Unseen

Stack (top at right): (empty)

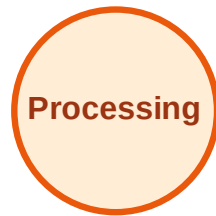
Visited: (none)



DFS Traversal

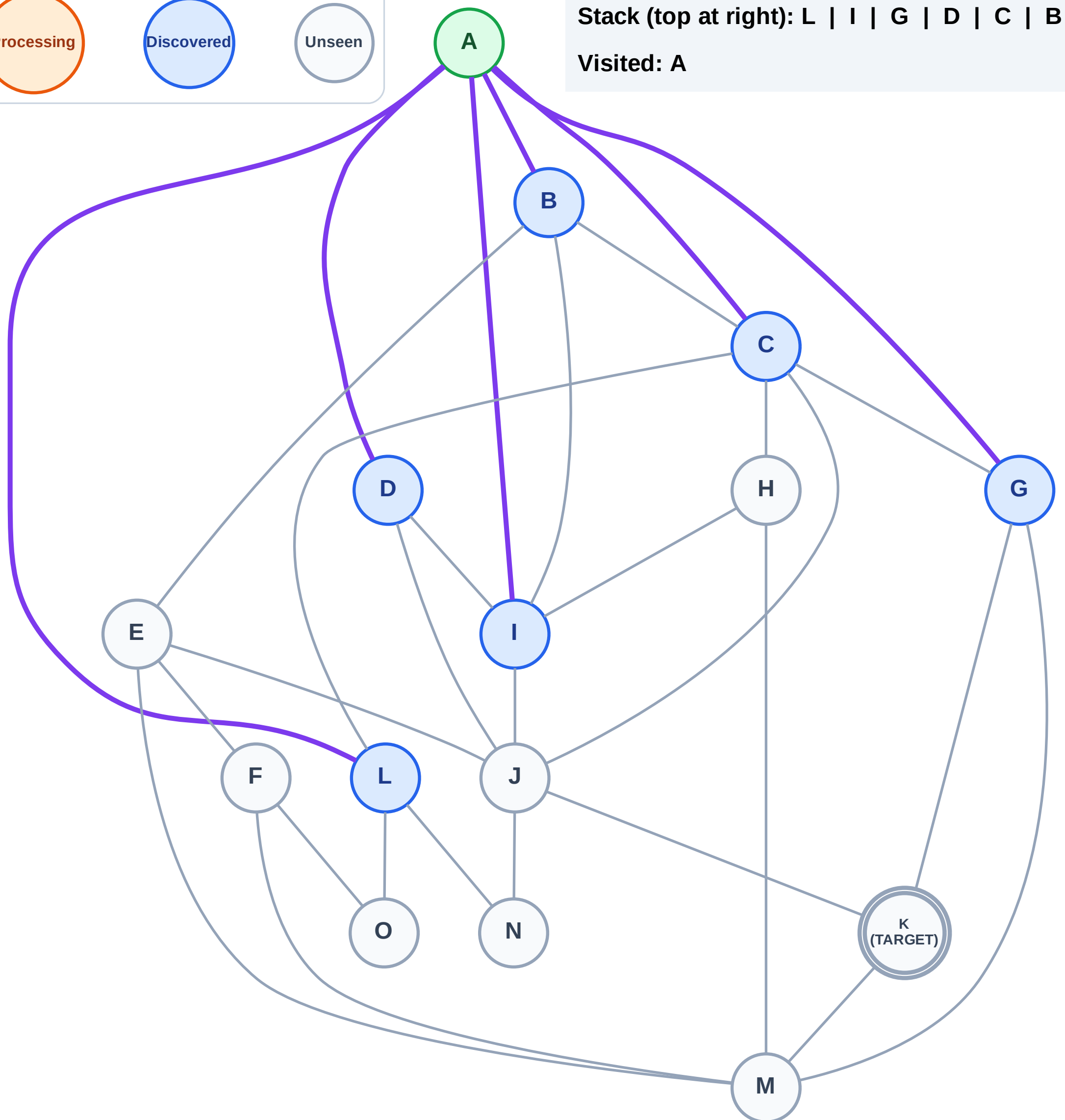
Step 3: Discover L, I, G, D, C, B from A

Legend



Stack (top at right): L | I | G | D | C | B

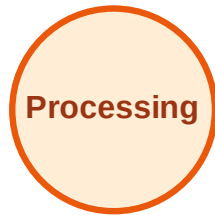
Visited: A



DFS Traversal

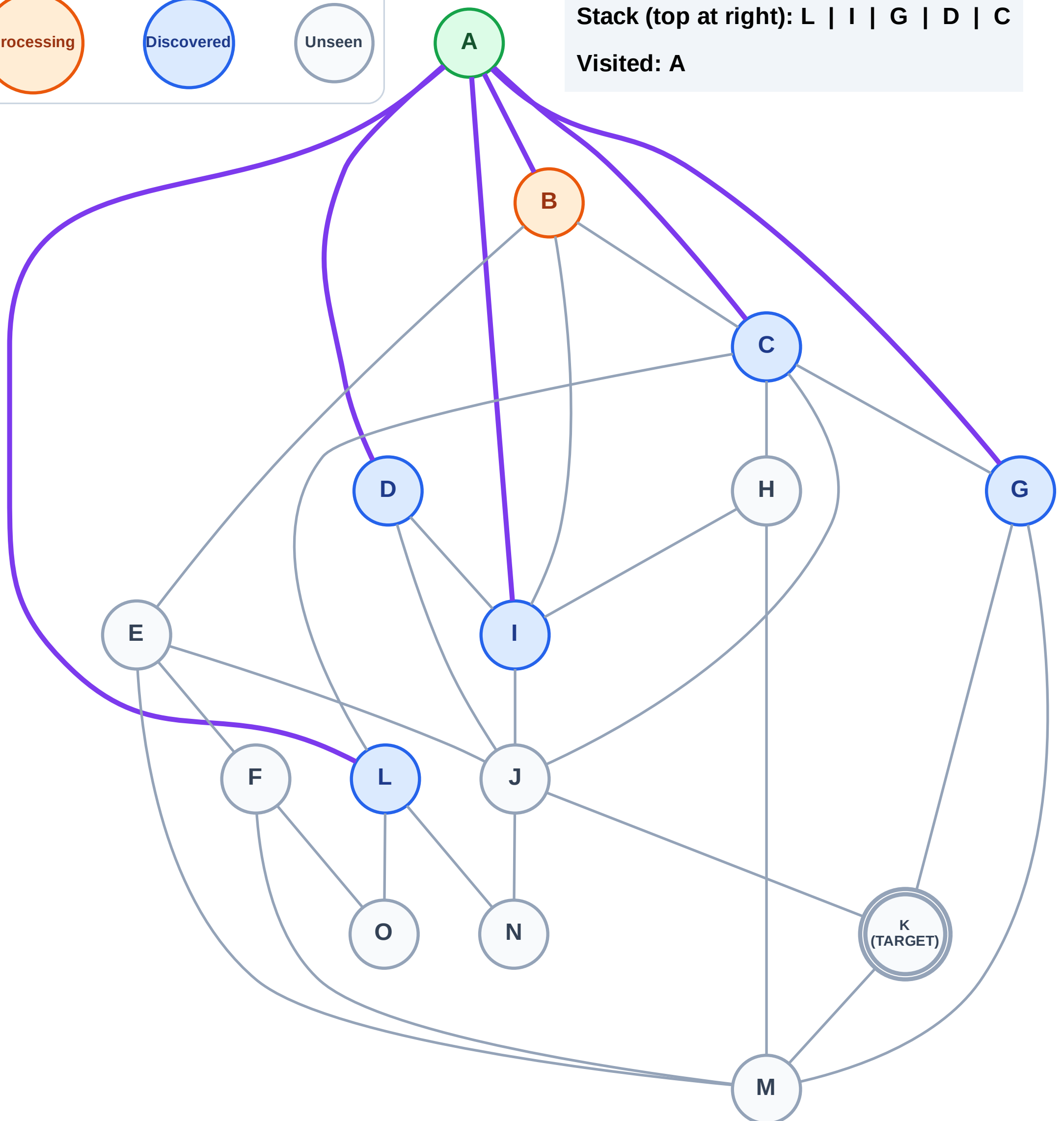
Step 4: Process node B

Legend



Stack (top at right): L | I | G | D | C

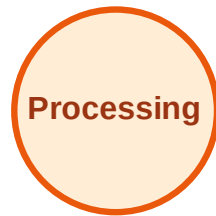
Visited: A



DFS Traversal

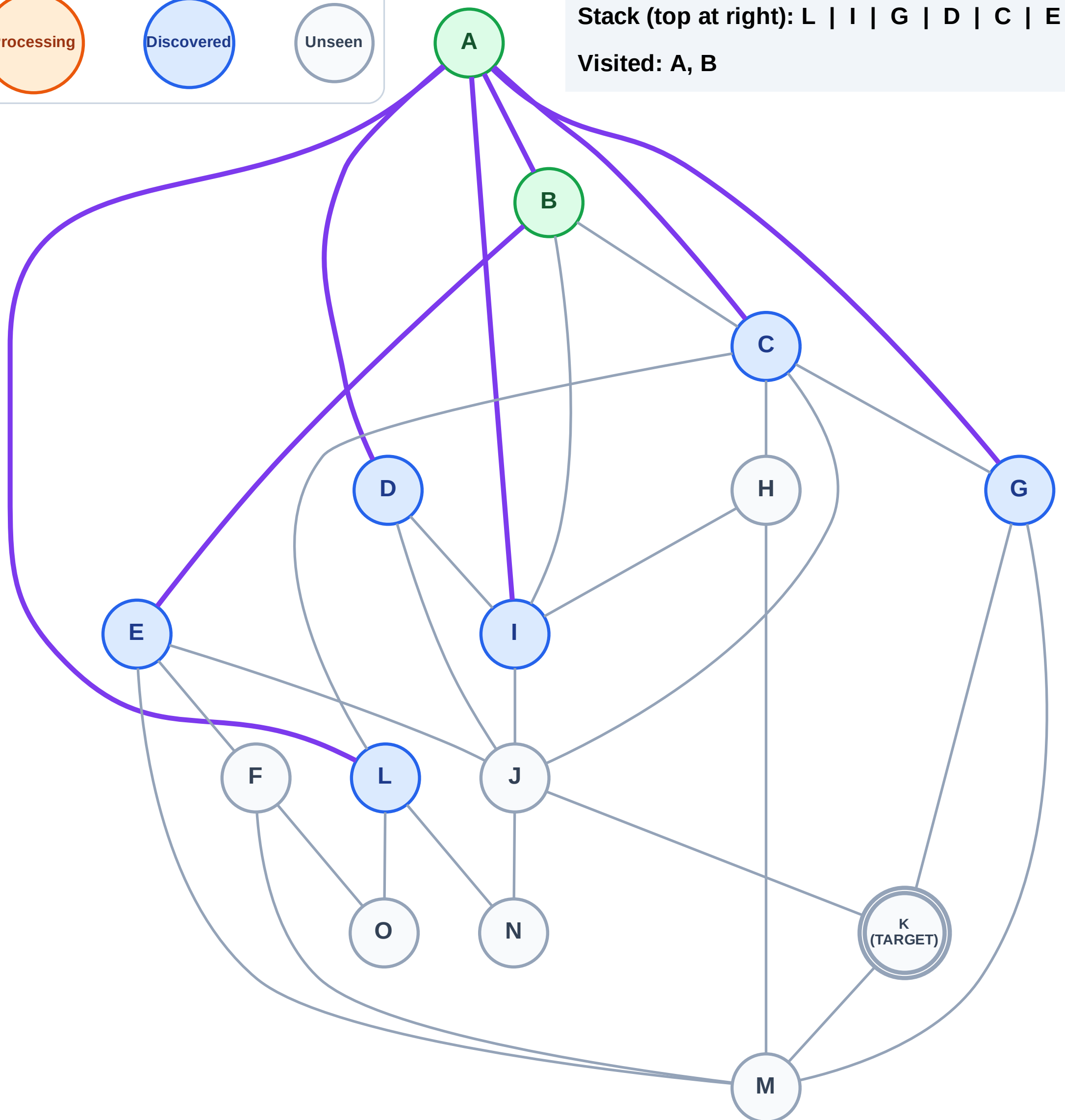
Step 5: Discover E from B

Legend



Stack (top at right): L | I | G | D | C | E

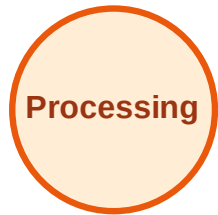
Visited: A, B



DFS Traversal

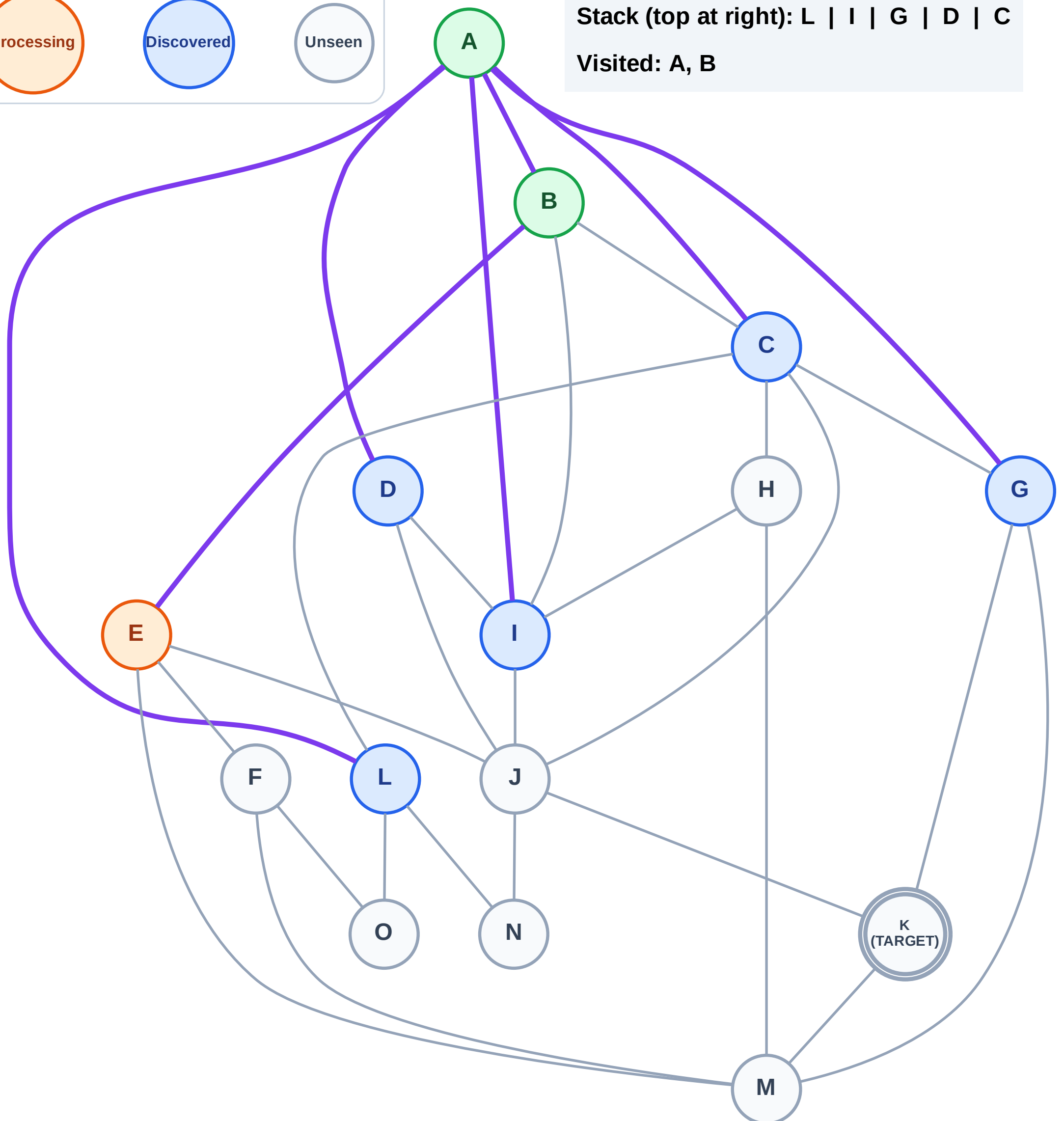
Step 6: Process node E

Legend



Stack (top at right): L | I | G | D | C

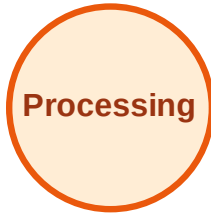
Visited: A, B



DFS Traversal

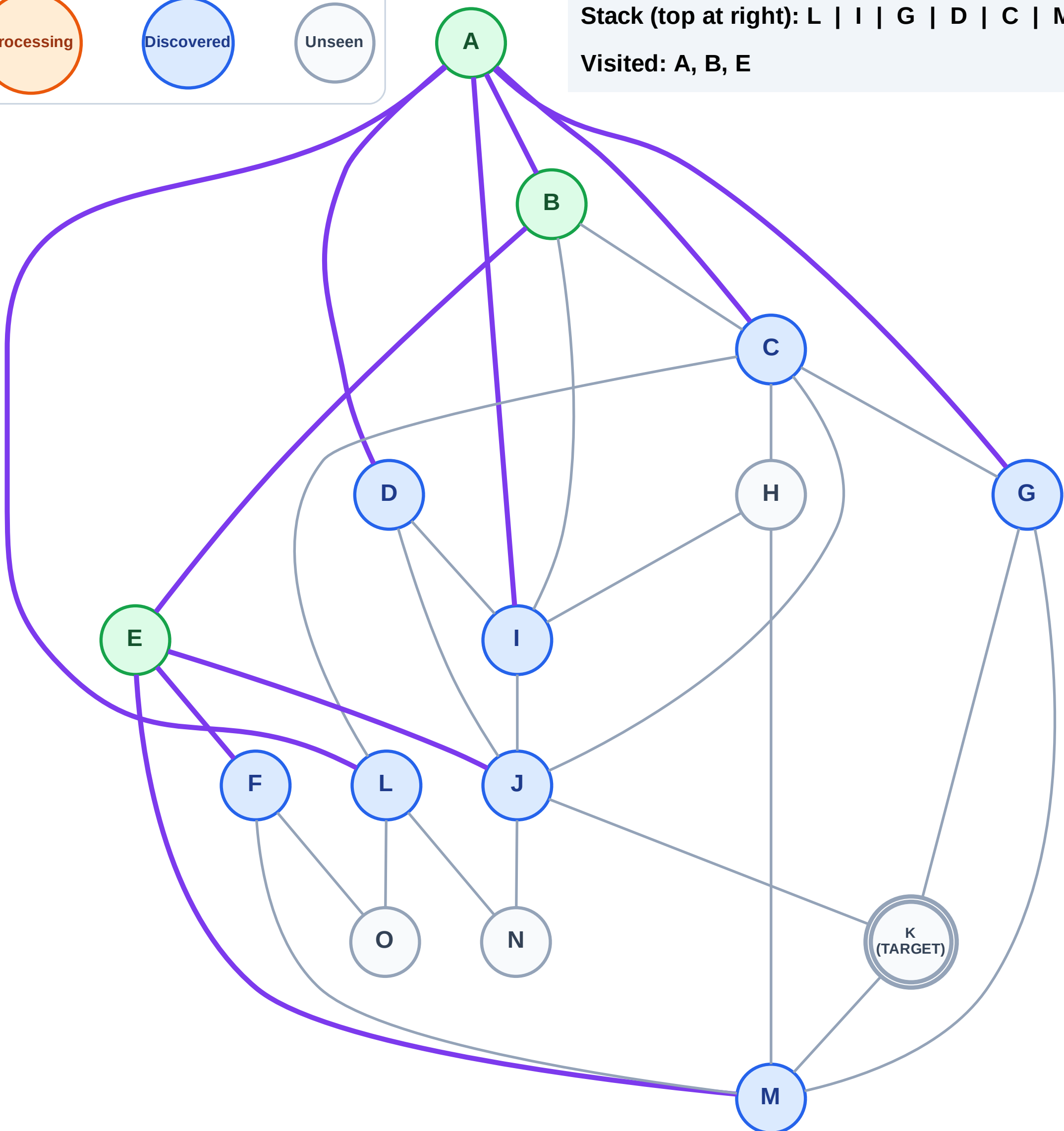
Step 7: Discover M, J, F from E

Legend



Stack (top at right): L | I | G | D | C | M | J | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

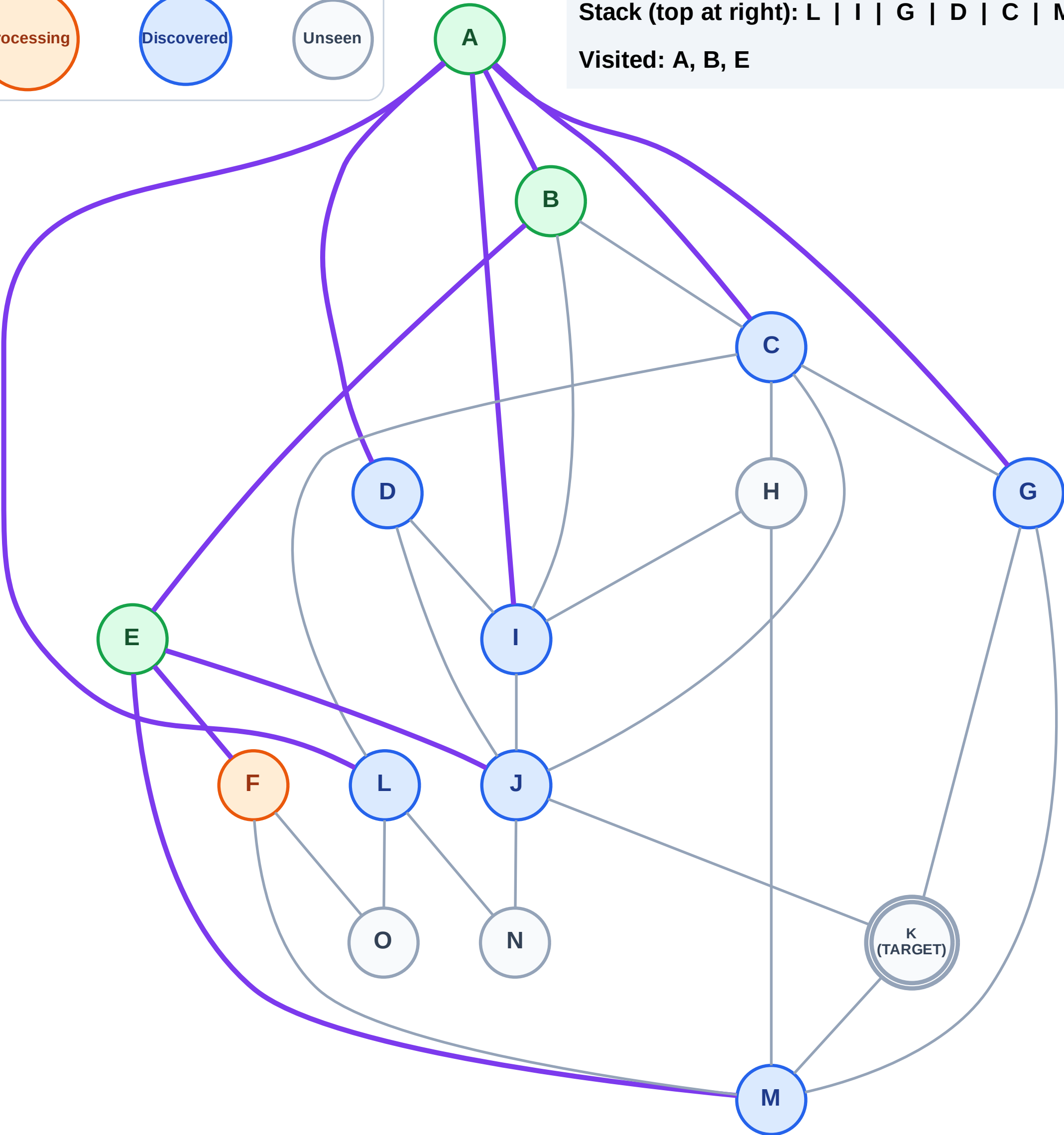
Processing

Discovered

Unseen

Stack (top at right): L | I | G | D | C | M | J

Visited: A, B, E



DFS Traversal

Step 9: Discover O from F

Legend

Complete

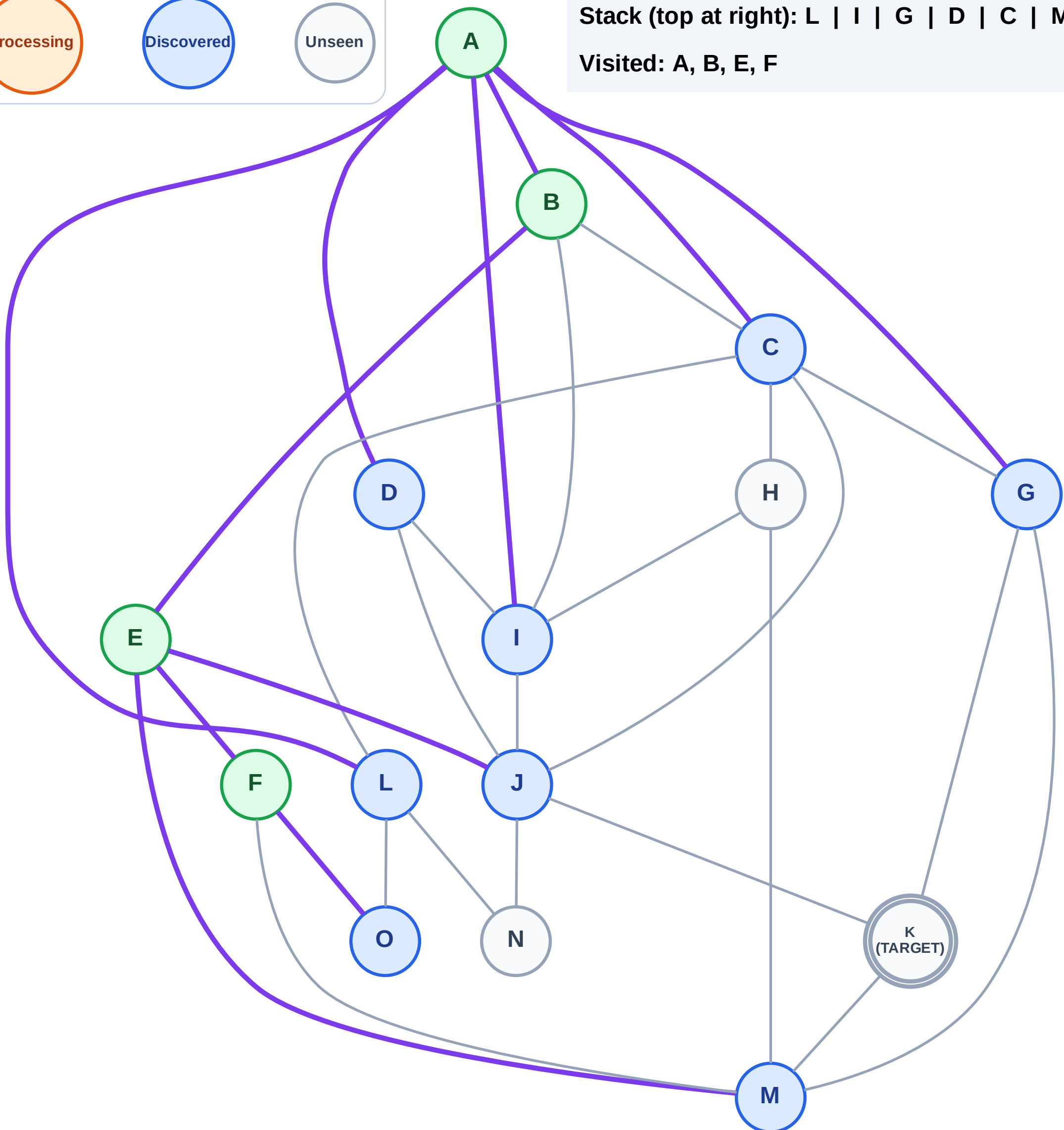
Processing

Discovered

Unseen

Stack (top at right): L | I | G | D | C | M | J | O

Visited: A, B, E, F



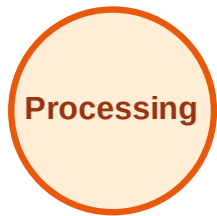
DFS Traversal

Step 10: Process node O

Legend



Complete



Processing



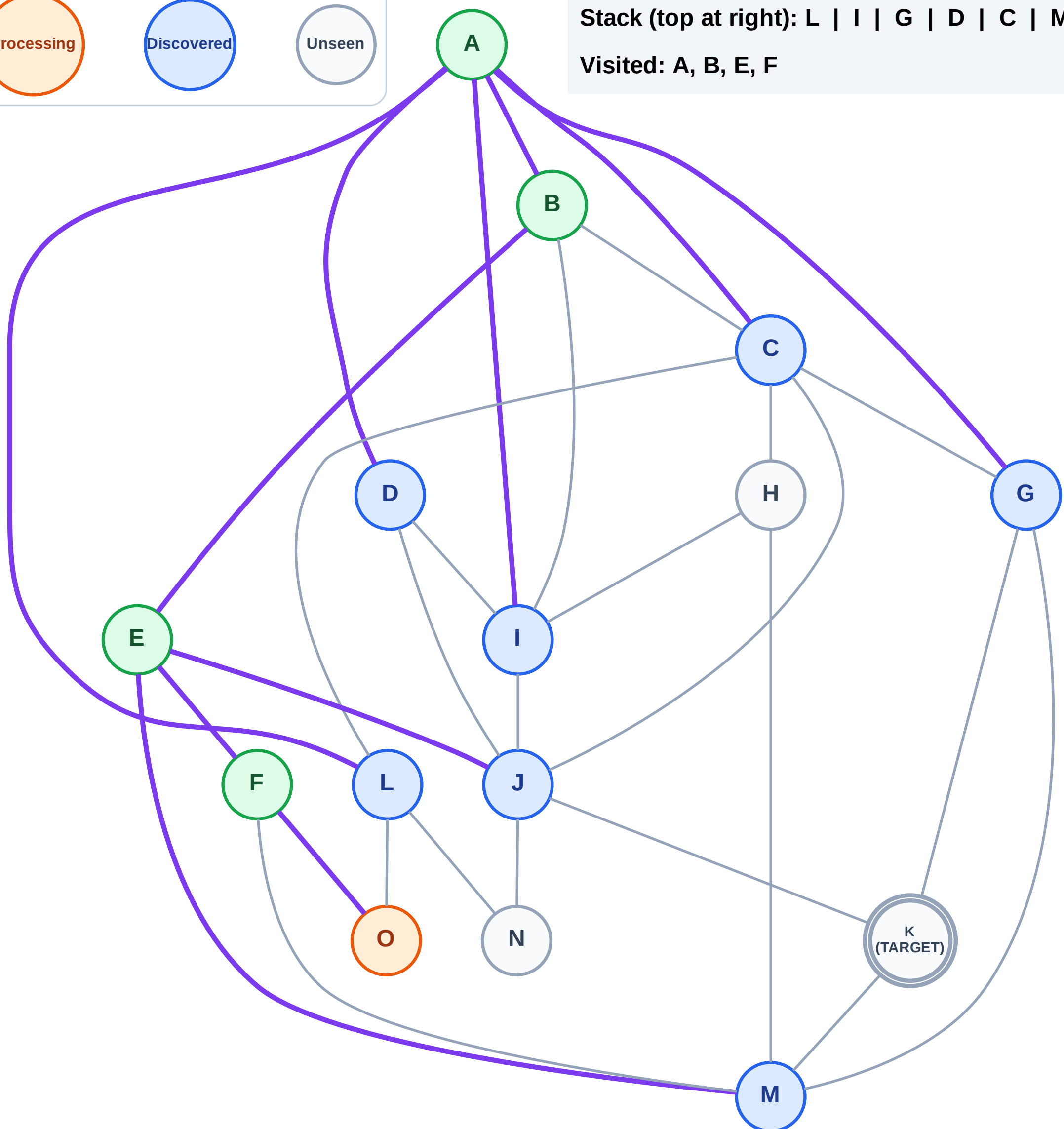
Discovered



Unseen

Stack (top at right): L | I | G | D | C | M | J

Visited: A, B, E, F



DFS Traversal

Step 11: No new neighbors from O

Legend



Complete



Processing



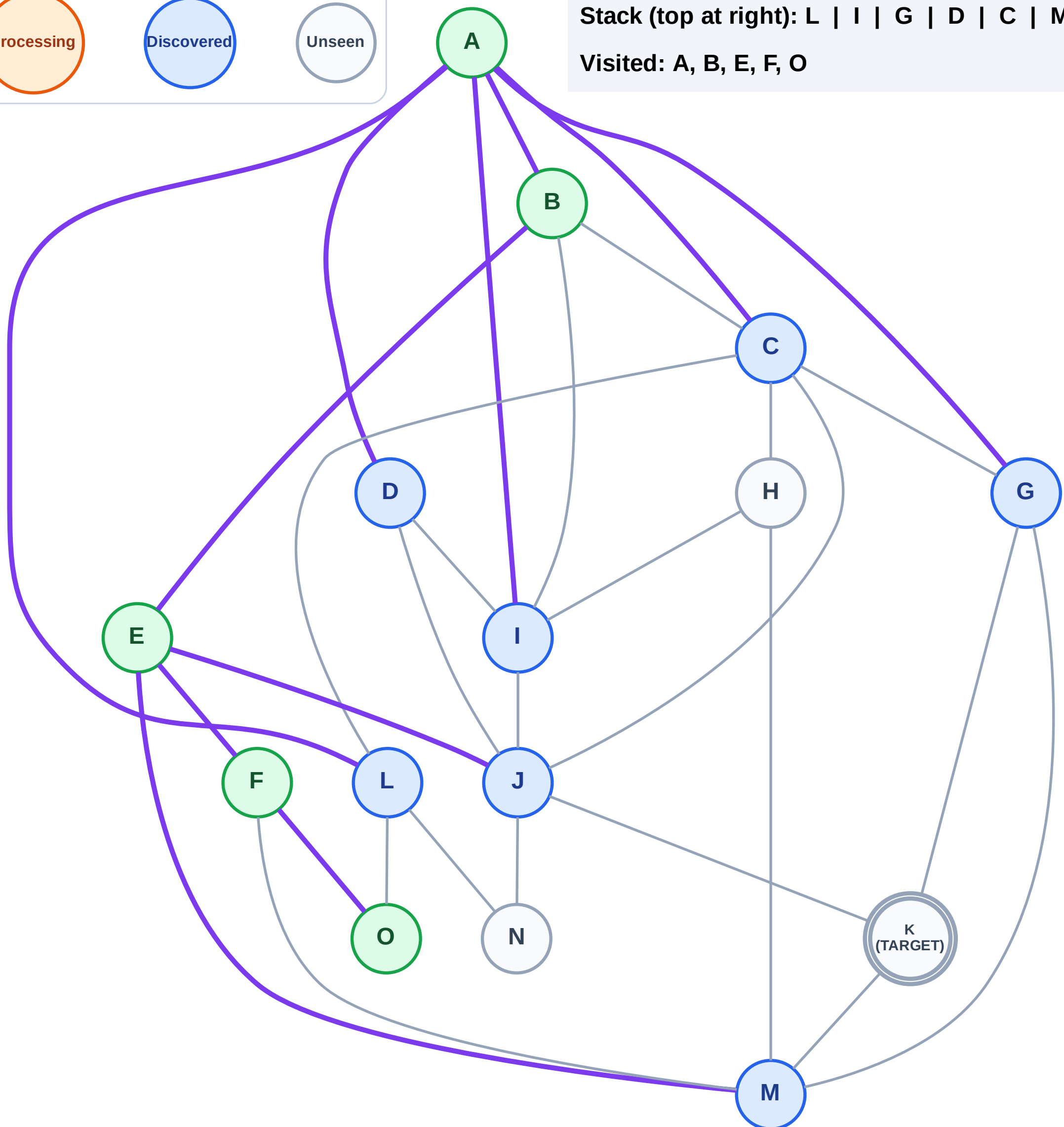
Discovered



Unseen

Stack (top at right): L | I | G | D | C | M | J

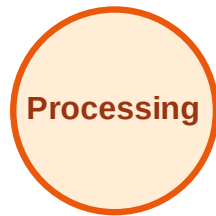
Visited: A, B, E, F, O



DFS Traversal

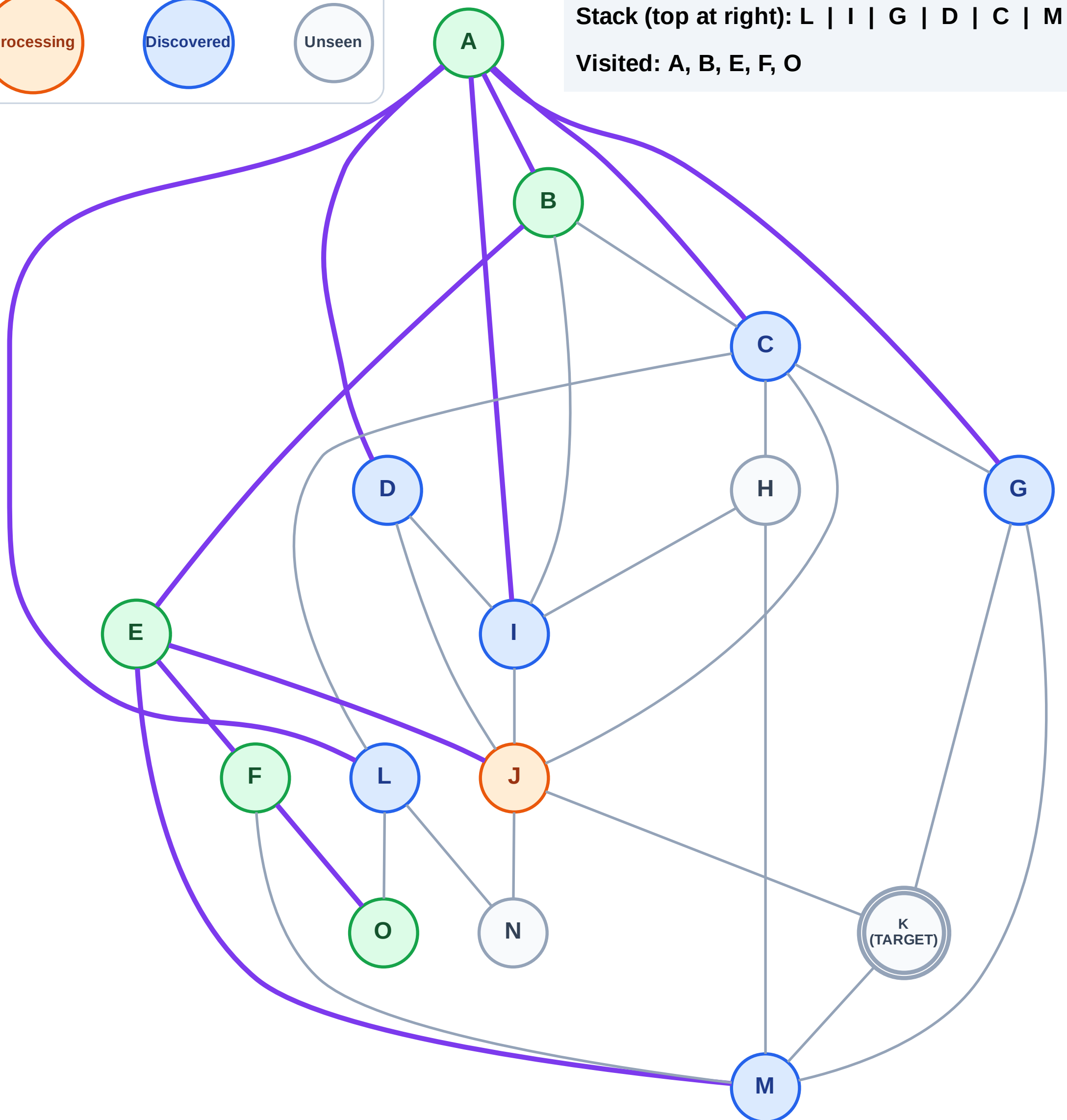
Step 12: Process node J

Legend



Stack (top at right): L | I | G | D | C | M

Visited: A, B, E, F, O



DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → E → J → K

Legend



Stack (top at right): L | I | G | D | C | M | N

Visited: A, B, E, F, J, K, O

